

KWANELE NTSHANGASE

0789325315 · kwanelerho69@gmail.com · linkedin.com/in/kwanele-ntshangase-abab7037b · github.com/dev-k99 · Portfolio

Junior .NET Fullstack Developer — ASP.NET Core 8 · Clean Architecture · Azure · React | Published on GitHub Marketplace · 8+ live deployed systems · SA regulatory expertise: POPIA compliance & SAPS/ITAC-compliant workflow digitisation

EDUCATION

Nelson Mandela University

— Dec 2025

Diploma in Information Technology

Khulumezulu High School

— Nov 2021

National Senior Certificate

SKILLS

.NET Stack: ASP.NET Core 8, EF Core 8, SignalR, Razor MVC, JWT, xUnit, Moq, Clean Architecture, FluentValidation

Frontend: React 18/19, TypeScript, JavaScript, Tailwind CSS, Vite, Zustand, TanStack Query v5

Languages: C#, TypeScript, JavaScript, Python, SQL

Databases: SQL Server, PostgreSQL, Azure SQL, Supabase, SQLite

Cloud & DevOps: Azure App Service, GitHub Actions, Docker, Vercel, Render, OIDC Workload Identity Federation

AI & Tooling: LangGraph, FastAPI, ChromaDB, RAGAS, Power Automate, Zapier, n8n, Power BI, Excel

Soft Skills: Critical Thinking, Problem-Solving, Effective Communication, Leadership

Languages: English, IsiZulu, IsiXhosa

PROJECTS

ScrapFlow SA | Full-Stack Scrapyard Management System

— March 2026

- Built and deployed a full-stack SAPS/ITAC-compliant scrapyard management platform using ASP.NET Core 8 (Clean Architecture, EF Core), React 18, and PostgreSQL, implementing a 6-step digitised inbound ticket workflow (vehicle arrival → weighing → grading → tare → payment → completion) that eliminates paper-based processing across multiple yard sites.
- Engineered real-time inventory tracking using ASP.NET Core SignalR WebSockets, broadcasting live stock updates to all connected React clients on ticket completion — removing the need for manual page refreshes and enabling accurate multi-site inventory visibility without polling.
- Implemented a 5-role RBAC system (Owner, Operator, Cashier, Gate, Auditor) secured with JWT bearer authentication and ASP.NET Core authorization policies; deployed to Render + Vercel with GitHub Actions CI/CD; validated core business logic with an xUnit 2.6 + Moq 4.20 test suite.
- Designed a webhook automation engine using HMAC-SHA256 signed HTTP payloads integrating with Zapier and n8n to trigger supplier payment receipts and manager alerts on business events — with a React/Zustand Owner-only UI for creating, testing, and toggling webhook subscriptions without code changes.

Eskhawini LibraryOS | Full-Stack Library Management System

— Jan 2026

- Architected and deployed a production-grade REST API using ASP.NET Core 8 and Clean Architecture (3-layer: API / Core / Infrastructure), exposing 35+ endpoints across 7 controllers with JWT authentication, refresh token rotation, role-based access control (Admin, Librarian, Patron), and fixed-window rate limiting — hosted on Azure App Service with PostgreSQL 17 (Azure Flexible Server) and automated EF Core migrations on startup.
- Built a real-time notification system using SignalR WebSockets with group-based routing and a daily background service that probes active loans for due-date alerts; resolved a JWT refresh race condition on the frontend using a subscriber queue pattern, eliminating duplicate refresh calls across parallel Axios requests.
- Developed a responsive React 19 SPA (Vite, Tailwind CSS) with Recharts analytics dashboards, QR code loan management (ZXing.Net), voice search via the Web Speech API, and a multi-step password reset flow backed by database-persisted cryptographically secure 6-digit tokens (RandomNumberGenerator) — deployed to Vercel with automatic CI/CD on every GitHub push.
- Eliminated N+1 database queries by refactoring repository interfaces to use eager-loaded GetAllAsync() with .Include(); implemented a GlobalExceptionHandler returning structured ProblemDetails JSON with correlation IDs; added PostgreSQL health check endpoints (/health/ready, /health/live) used as Azure platform liveness and readiness probes.

WardCare+ | ASP.NET Core 8 MVC — Team Project, Final Year

— Oct 2025

- Solely owned the cloud deployment pipeline for a 4-person team project: provisioned Azure App Service and Azure SQL Database, configured GitHub Actions CI/CD with OIDC workload identity federation (zero stored credentials), and automated EF Core migrations on startup — reducing every release to a single push to main.
- Collaborated in a team of 4 to architect and deliver a production-grade ASP.NET Core 8 MVC application with 7 RBAC roles, policy-based authorization, FluentValidation, and a service-layer pattern — applying Git branch workflows throughout to enable parallel development across subsystems.
- Designed and delivered the Doctor–Patient subsystem end-to-end: C# MVC controllers, Razor views, EF Core 8 queries, Bootstrap 5 UI, and in-memory PDF generation (QuestPDF) for downloadable clinical patient folders.
- Implemented a POPIA-compliant audit trail using an EF Core SaveChangesInterceptor capturing before/after JSON snapshots on every INSERT, UPDATE, and DELETE across 25+ entities with a soft-delete pattern on all tables — validated with an xUnit + Moq test suite covering the interceptor contract and service-layer logic.

RaptorReview AI | GitHub Actions — Published on GitHub Marketplace

— 2025

- Published a GitHub Action on the GitHub Marketplace that automatically reviews every pull request using Groq LLM inference, posting line-specific security, performance, and readability comments directly in the PR diff interface — installable by any repository in one workflow YAML step with zero hosted infrastructure or paid API key required.
- Engineered diff-hunk-only context segmentation using tiktoken to keep token count proportional to change size (not file size), preventing context window overflows on real codebases regardless of file length.

ADDITIONAL PROJECTS

DocLens (Python, FastAPI, ChromaDB, RAGAS) — PDF RAG system with hybrid BM25 + dense vector retrieval and cross-encoder reranking; +25–31% improvement across all four RAGAS metrics vs cosine-similarity baseline. github.com/dev-k99/RAG-Assistant

MemOS (Python, LangGraph, PostgreSQL, LangSmith) — Agentic AI assistant with custom LangGraph StateGraph, persistent cross-session memory, and full LangSmith observability. github.com/dev-k99/MemOS

PulseTrace (Python, Streamlit, Docker) — Free AI agent monitoring dashboard with OpenTelemetry-style traces and ZAR cost tracking; deployed on HuggingFace Spaces. github.com/dev-k99/pulsetrace

Kova (React 18, TypeScript, Zustand, TanStack Query v5, Paystack) — Production e-commerce storefront with ZAR checkout, admin dashboard, URL-synced filters, and aria-live accessibility. github.com/dev-k99/Kova

CERTIFICATIONS & AWARDS

Angular & ASP.NET Core Web API — Udemy, Oct 2025

AI Fluency Framework & Foundations — Anthropic, Mar 2026

Introduction to Cybersecurity — Cisco Networking Academy, Oct 2023

Data Analyst Bootcamp: Python, Excel, Power BI & SQL — Udemy, Oct 2025

C# Programming Award — Nelson Mandela University, Apr 2024

Student Merit Awards (2024 & 2025) — Nelson Mandela University